بسم الله الرحمن الرحيم

Discrete Mathematics

CHAPTER 1

Propositional Logic

RABNAWAZ MALLAH
Lecturer in Mathematics
Aror University of Art, Architecture, Design and Heritage Sukkur Sindh
Rabnawaz.faculty@aror.edu.pk
0308-8484215

Introduction to Discrete Mathematics

Target Audience:

» Students who are preparing for GATE and other competitive exams.
» Students who want to learn competitive programming.
» College going students who have Discrete Mathematics in their syllabus.
» Everyone who wants to learn Discrete Mathematics as a whole or may be a small subset of this subject.

Why Study Discrete Mathematics?

(1) It develops your mathematical thinking.
(2) Improves your problem solving ability.
(3) If you are computer science student, then no need to go anywhere else because Discrete Mathematics is for you.
Discrete Mathematics is important to survive in subjects like: compiler design, databases, computer security, operating system, automata theory etc.
(4) Many problems can be solve using Discrete Mathematics:

For Example:

- Sorting the list of integers.
- Finding the shortest path from your home to your friend's home.
- Drawing a graph with two conditions:
    1. You are not allowed to lift your pen.
    2. You are not allowed to repeat edges.

Try drawing this graph.

- How many different combinations of passwords are possible with just 8 alphanumeric characters.
- Encrypt a message and deliver it to your friend and you don't want anybody to read that message except your friend.

What is Discrete Mathematics?

» Discrete Mathematics is the study of discrete objects.

Discrete means: "distinct or not connected".

» It is not a branch of Mathematics. It is rather a description of set of branches that have one common property - that they are "discrete" and not "continuous".

Discrete vs Continuous

» The whole world of Mathematics is divided in two realms:

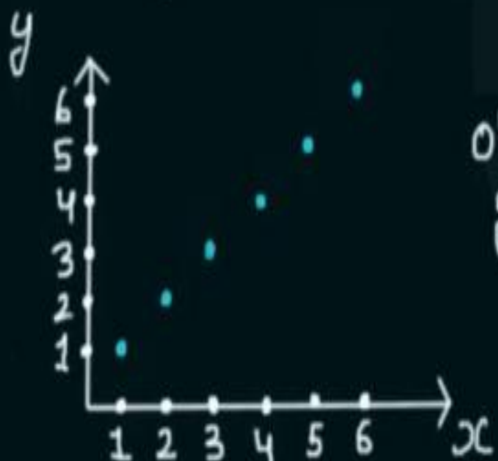
1. Discrete 2. Continuous

Discrete Objects:

1. Natural Numbers are discrete.

For Example: 1, 2, 3, 4, 5...

$y = x \quad x \in \underline{N} \quad y \in \underline{N}$



observe the gaps.

Continuous Objects:

1. Real numbers are continuous

For Example: between 0 and 1 -

0, 0.001..., 0.000001..., 0.10001..., and so on

$y = x \quad x \in \mathbb{R} \quad y \in \mathbb{R}$

Continuous line

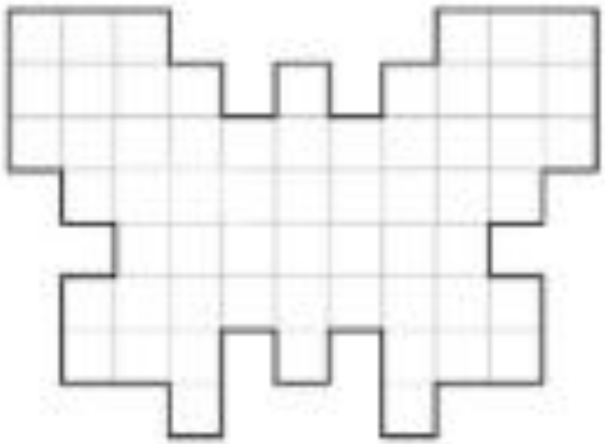
No gaps.

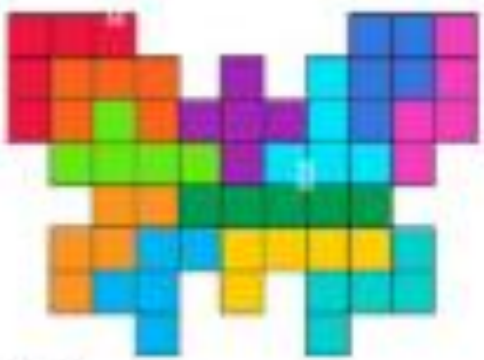
2. Digital clock is "discrete" in nature.
- there is no continuous time,
- Transition from one time to another is sharp.

2. Analog clock is continuous in nature.
- In Analog clock - hour, minute and second hands move smoothly over time.

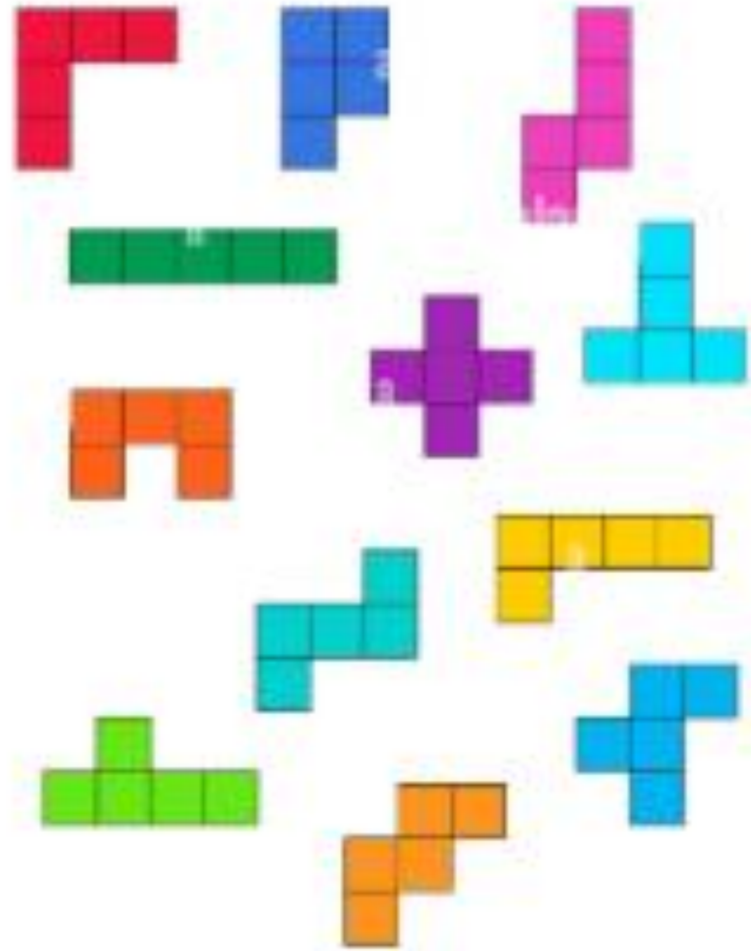
Consider the clock where minute hand, hour hand and second hand sweeps around the time smoothly

Complete the figure from the details





ANSWER



Syllabus:

1. Propositional Logic and First Order Logic
2. Set theory
3. Relations
4. Functions
5. Partial Orders and Lattices
6. Combinatorics
7. Graph Theory
8. Group Theory

Motivation and Introduction to Propositional Logic

The phrase propositional logic is composed of two words:

Proposition | Logic

What is Logic?

- Logic is the science of reasoning.
  It helps us to understand and reason about different mathematical statements.

- With rules of logic, we would be able to think about mathematical statements and finally we would be able to prove or disprove those mathematical statements precisely.

For example:

Rules of logic enable us to reason about the mathematical statements like:
"For every positive integer n, the sum of positive integers not exceeding n is n(n+1)/2".
Either it is a valid mathematical argument or it is an invalid argument.

Purpose of logic is to construct valid arguments (or proofs).
Once we prove a mathematical statement is TRUE then we call it a Theorem.
and this is the basis of whole mathematics.

Example 2:

Famous Knights and Knaves Puzzle:

"In an Island there are two kinds of inhabitants, Knights, who always tell the truth, and their opposites, Knaves, who always lie. You encounter two people Alice and Bob.
Who are Alice and Bob if Alice says "Bob is a Knight" and Bob says "The two of us are opposite types?"

These types of problems can be very easily solved using an area of logic called propositional logic.

Propositional Logic:

What is proposition?

Proposition is a declarative sentence (a sentence that is declaring a fact or stating an argument) which can be either TRUE or FALSE but cannot be both.

For example:

(1) Delhi is the capital of India. **Islamabad is the capital of Pakistan.**

(2) Water Froze this morning.

(3) $1 + 1 = 2$.

Sentences which are not propositions:

(1) What time is it?

(2) $x + 1 = 2$ (can be both TRUE or FALSE)

(3) Send us your resume before 11 PM.

(4) I request you to please allow me a day off.

(5) Fetch my umbrella!

Propositions (Statements that are either true or false):

1. $2 + 2 = 4$ (True)
2. $5 > 10$ (False)
3. "The Eiffel Tower is in Paris." (True)
4. "Water boils at 100°C at sea level." (True)
5. $x + 3 = 7$, when $x = 4$ (True)
6. "The moon is made of cheese." (False)
7. "Every prime number greater than 2 is odd." (True)
8. $3 \times 3 = 10$ (False)
9. "If it rains, then the ground gets wet." (True under normal conditions)
10. "All swans are white." (False if counterexamples exist)

Non-Propositions (Statements that are not definitively true or false):

1. "What time is it?" (A question)
2. "Close the door." (A command)
3. "x + 5 = 10" (Depends on the value of x)
4. "Please pass the salt." (A polite request)
5. "Is the sky blue?" (A question)
6. "Let's go to the park." (A suggestion)
7. "This sentence is false." (A paradox)
8. "I wish it were sunny." (An expression of desire)
9. "Run faster!" (A command)
10. "He is taller than her." (Ambiguous unless "he" and "her" are defined)

EXAMPLE 2 Consider the following sentences.

1. What time is it?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$.

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false. Note that each of sentences 3 and 4 can be turned into a proposition if we assign values to the variables. We will also discuss other ways to turn sentences such as these into propositions in Section 1.4. ◀

Exercises

1. Which of these sentences are propositions? What are the truth values of those that are propositions?

a) Boston is the capital of Massachusetts.
b) Miami is the capital of Florida.
c) $2 + 3 = 5$.
d) $5 + 7 = 10$.
e) $x + 2 = 11$.
f) Answer this question.

2. Which of these are propositions? What are the truth values of those that are propositions?

a) Do not pass go.
b) What time is it?
c) There are no black flies in Maine.
d) $4 + x = 5$.
e) The moon is made of green cheese.
f) $2^n \geq 100$.

3. What is the negation of each of these propositions?

a) Mei has an MP3 player.
b) There is no pollution in New Jersey.
c) $2 + 1 = 3$.
d) The summer in Maine is hot and sunny.

4. What is the negation of each of these propositions?

a) Jennifer and Teja are friends.
b) There are 13 items in a baker's dozen.
c) Abby sent more than 100 text messages every day.
d) 121 is a perfect square.

Homework Problem:

Which of the following is/are proposition(s).
Also if you find the proposition, state whether it is True or False:

1. London is in Denmark.
2. Do your Homework.
3. India wins the match by 2 runs.
4. x is an even number.
5. 5 is an odd number.
6. Rahul.
7. $5 + 7 = 10$
8. The moon is made of cheese.
9. The only odd prime number is 2.
10. God bless you!

Propositional logic, Propositional Variables and Compound Propositions

Propositional Logic

area of logic that studies ways of joining and/or modifying propositions to form more complicated propositions and it also studies the logical relationships and properties derived from these combined/altered propositions.

What does this definition really mean?

Statement 1 - "Adam is good in playing football"

Statement 2 - "Adam is good in playing football and this time he is representing his college at National level."

"Adam is good in playing football and this time he is representing his college at National level."

proposition 1

joining two propositions with logical connective

proposition 2

Why do we need compound propositions?

because most of the mathematical statements are constructed by combining one or more than one propositions. As simple as that!

Propositional Variables:

Which one of the following is convenient to express:

"Adam is good in playing football and this time he is representing his college at National level."

OR

if p = Adam is good in playing football

q = this time he is representing his college at National level.

$p \wedge q \longleftarrow$ p and q

Definition: Variables that are used to represent propositions are called Propositional Variables.

Negation, Conjunction and Disjunction

There are 6 logical operators that we will focus on:

1. Negation
2. Conjunction
3. Disjunction
4. Exclusive OR
5. Implication
6. Biconditional

our focus in this lecture.

Negation, Conjunction and Disjunction

Negation:

Let p be a proposition. $\neg p$ is called negation of p which simply states that

"It is not the case that p"

if p is true then $\neg p$ is false. if p is false then $\neg p$ is true.

Example: let p be a proposition - "Adam and Eve lived together for many years."

then $\neg p$ will be :

"It is not the case that Adam and Eve lived together for many years."

OR

"Adam and Eve haven't lived together for many years."

p	$\neg p$
T	F
F	T

Conjunction:

Let p and q be two propositions. Conjunction of p and q is denoted by $p \wedge q$
When both p and q are true then only the compound proposition $p \wedge q$ is true.

Example: "12 is divisible by 3 and 3 is a prime number"

T T = T

Important Note: Sometimes we use 'but' instead of 'and'.

Example: "12 is divisible by 3 but 3 is a prime number"

↓

and

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

CONJUNCTION

Let p and q be propositions. The conjunction of p and q, denoted by p ∧ q, is the proposition "p and q." The conjunction p ∧ q is true when both p and q are true and is false otherwise.

EXAMPLE

P : Islamabad is the capital of Pakistan.

q : The currency of Pakistan is dollar.

$p^\wedge q$ **: Islamabad is the capital of Pakistan and his currency is a dollar.**

$p^\wedge - q$ **: Islamabad is the capital of Pakistan and his currency is not a dollar.**

$-p^\wedge q$ **: Islamabad is not capital of Pakistan and his currency is a dollar.**

$-p^\wedge - q$ **: Islamabad is not capital of Pakistan and his currency is not a dollar.**

EXAMPLE

P : Islamabad is the capital of Pakistan.

q : The currency of Pakistan is dollar.

$p^\wedge q$: Islamabad is the capital of Pakistan OR his currency is a dollar.

$p^\wedge - q$: Islamabad is the capital of Pakistan OR his currency is not a dollar.

$-p^\wedge q$: Islamabad is not capital of Pakistan OR his currency is a dollar.

$-p^\wedge - q$: Islamabad is not capital of Pakistan OR his currency is not a dollar.

p ∧ q

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction:

Let p and q be propositions. Disjunction of p and q is denoted by $p \vee q$

When both p and q are false then only the compound proposition $p \vee q$ is false.

Example: "16 - 4 = 10 or 4 is an even number"

X ✓

F T = T

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

DISJUNCTION (∨) or INCLUSIVE OR

Let p and q be propositions. The disjunction of p and q, denoted by p ∨ q, is the proposition "p or q." The disjunction p ∨ q is false when both p and q are false and is true otherwise.

EXAMPLE

P : Islamabad is the capital of Pakistan.

q : The currency of Pakistan is dollar.

pVq **: Islamabad is the capital of Pakistan OR his currency is a dollar.**

$pV-q$ **: Islamabad is the capital of Pakistan OR his currency is not a dollar.**

$-pVq$ **: Islamabad is not capital of Pakistan OR his currency is a dollar.**

$-pV-q$ **: Islamabad is not capital of Pakistan OR his currency is not a dollar.**

EXAMPLE

P : Islamabad is the capital of Pakistan.

q : The currency of Pakistan is dollar.

$p \vee q$: Islamabad is the capital of Pakistan **OR** his currency is a dollar.

$p \vee -q$: Islamabad is the capital of Pakistan **OR** his currency is not a dollar.

$-p \vee q$: Islamabad is not capital of Pakistan **OR** his currency is a dollar.

$-p \vee -q$: Islamabad is not capital of Pakistan **OR** his currency is not a dollar.

TRUTH TABLE FOR

$p \vee q$

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

According to problem

p	q	$p \vee q$
T	F	T
T	T	T
F	T	T
F	F	F

Logical Operator - Exclusive OR

Consider the following statement:

"In order to get a job in this multinational company, experience with C++ or Java is mandatory."

p

q

Inclusive OR (Disjunction).

Experience with C++ ✓
" " Java ✓
both ✓

p or q or both

Definition: Let p and q be two propositions. The exclusive OR of p and q (denoted by $p \oplus q$) is a proposition that simply means exactly one of p and q will be true but both cannot be true.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

EXCLUSIVE OR

Let p and q be propositions. The exclusive or of p and q, denoted by p ⊕ q, is the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

P : Parrot can fly.

Q : Karachi is the capital of Sindh.

Implication - Definition and Examples

Definition:

Let p and q be propositions. The proposition "if p then q" denoted by p -> q is called implication or conditional statement.

Truth Table

p	q	p -> q
T	T	T
T	F	F
F	T	T
F	F	T

p is called hypothesis (or premise) and q is called conclusion (or consequence)

IMPLICATION

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition "if p, then q." The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

WAY OF WRITTING

"if p, then q"	"p implies q"
"if p, q"	"p only if q"
"p is sufficient for q"	"a sufficient condition for q is p"
"q if p"	"q whenever p" "q when p"
"q is necessary for p"	"a necessary condition for p is q"
"q follows from p"	"q unless ¬p"

EXAMPLE

Let p be the statement "Maria learns discrete mathematics" and q the statement "Maria will find a good job."

Express the statement $p \rightarrow q$ as a statement in English.

SOLUTION

"if p, then q"

"If Maria learns discrete mathematics, then she will find a good job.

"q only/when if p"

"Maria will find a good job when she learns discrete mathematics."

"q unless ¬p"

"Maria will find a good job unless she does not learn discrete mathematics."

Example 1:

"If you try hard for your exam, then you will succeed".

p = you tried hard for your exam.

q = you succeed

Case 1: "You tried hard for your exam" and "you succeed"

p = True q = True

Compound proposition p -> q is True

Case 2: "You tried hard for your exam" but "you failed"

p = True q = False

Compound proposition p -> q is False.

Example 1:

"If you try hard for your exam, then you will succeed".

p = you tried hard for your exam.

q = you succeed

Case 3: "You haven't tried hard for your exam" and "You succeeded"

p = False q = True

Compound proposition p -> q is True. Why?

because you can make the compound proposition false only when you satisfy the first condition itself i.e. p. If that itself not satisfied then we cannot make compound proposition False. Not False means True.

Case 4: "You haven't tried hard for your exam" and "You failed"

p = False q = False

Compound proposition p -> q is True. (Same reason as above)

"p only if q"

How "if p then q" and "p only if q" can be same?

Example: "I will stay at home only if I'm sick."

Let p = "I will stay at home" and let q = "I'm sick"

Above statement is of the form p only if q

According to the above statement, becoming sick is the necessary condition that will make you stay at home.

This means "if you're not sick then, you cannot stay at home at any cost."

In order to falsify the above statement, q must be FALSE and p must be TRUE i.e. you are not sick and you still stay at home.

Proof idea: truth value of p and q must be same in order to falsify the statement.

if p then q: "If I'll stay at home then I'm sick"

The only way to falsify the above statement is by making p TRUE and q FALSE. Therefore, p only if q is equivalent to if p then q

HENCE PROVED

"q is necessary for p"

Example: "Good food is neccessary to keep us alive"

According to the statement, if we won't have good food then, we'll soon die.

But is it the only factor to keep us alive? No.

Example: My friend died at the age of 16 and he has no shortage of good food.
(Even though he is having good food but still he died.)

Therefore, when we say A is necessary for B then falsity of A guarantees the falsity of B but <u>we cannot guarantee</u> the <u>truth</u> of B from the truth of A.

Similarly, when we say q is necessary for p, then we can only guarantee that when q is false then, p is definitely false.

"p is sufficient for q"

Example: "It is sufficient for you to travel by car in order to reach your destination on time."

Definitely, if you travel by car, you'll reach your destination on time. No doubt. but if you won't travel by car, does it mean you'll never reach your destination on time. May be by flight or other means of transport, you'll reach your destination much earlier.

Therefore, when we say that when A is sufficient for B then, truth of A guarantees the truth of B but <u>we cannot guarantee</u> the <u>falsity</u> of B from the falsity of A.

Similarly, when we say "p is sufficient for q", then we can only guarantee that when p is true then, q is definitely true.

Why p is not necessary for q?

p	q	p -> q
T	T	T
T	F	F
F	T	T
F	F	T

Proof by contradiction: Let say p is necessary for q.

if p is really necessary for q then if p is false then it is mandatory for q to be false in order to make the whole compound proposition p->q true. but it is not the case.

According to the truth table, it is not mandatory for q to be false when p is false. Hence "p is not necessary for q."

Why q is not sufficient for p?

p	q	p -> q
T	T	T
T	F	F
F	T	T
F	F	T

Proof by contradiction: Let say q is sufficient for p.

if q is really sufficient for p then if q is true then it is mandatory for p to be true in order to make the whole compound proposition p->q true. but it is not the case.

According to the truth table it is not mandatory for p to be true when q is true. Hence "q is not sufficient for p."

"q unless $\neg$p"

→ means 'except if'

Evolution steps:

"if p then q" (if p is true then q must be true)

↓

"q is true when p is true"

↓

"q is true except when p is false"

↓

"q is true unless p is false" = "q unless $\neg$p"

CONVERSE, CONTRAPOSITIVE, AND INVERSE

The proposition $p \to q$ then converse is $q \to p$

The proposition $p \to q$ then contrapostive is $\neg q \to \neg p$.

The proposition $p \to q$ then inverse is $\neg p \to \neg q$.

If it snows today, I will ski tomorrow.

a) **Converse: "I will ski tomorrow only if it snows today."**

b) **Contrapositive**: **"If I do not ski tomorrow, then it will not have snowed today."**

c) **Inverse: "If it does not snow today, then I will not ski tomorrow."**

Implication, Converse, Contrapositive and Inverse

Implication or conditional statement: $p \to q$

Converse - $q \to p$

Contrapositive - $\neg q \to \neg p$

Inverse - $\neg p \to \neg q$

Example: "If it rains today then, I will stay at home."

Converse - If I will stay at home then it rains today.

Contrapositive - If I will not stay at home, then it does not rain today.

Inverse - If it does not rain today then, I will not stay at home.

FACTS:

>> Implication and contrapositive both are equivalent.

>> converse and inverse both are equivalent.

>> Neither converse nor inverse is equivalent to Implication.

FACTS:

- » Implication and contrapositive both are equivalent.
- » converse and inverse both are equivalent.
- » Neither converse nor inverse is equivalent to Implication.

TRUTH TABLE

p	q	$\neg p$	$\neg q$	$p \to q$	$q \to p$	$\neg q \to \neg p$	$\neg p \to \neg q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

Biconditional Operator

Definition: let p and q be two propositions. The biconditional statement of the form $p \leftrightarrow q$ is the proposition "p if and only if q".
$p \leftrightarrow q$ is true whenever the truth values of p and q are same.

Truth Table

p	q	$p\leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

How "p if and only if q" make sense?

"p if and only if q" composed of two statements -
"p if q" and "p only if q".

"p only if q" = if p then q and "p if q" = if q then p

$$(p \rightarrow q) \wedge (q \rightarrow p) \equiv p \leftrightarrow q$$

Representations:

1. p is necessary and sufficient for q and vice versa"
2. if p then q, and conversely
3. p iff q

Example: let p be a proposition "You get promoted" and let q be a proposition "You have connections"

then $p \leftrightarrow q$ is the statement: "You get promoted if and only if you have connections."

Precedence of Logical Operators

Precedence of operators helps us to decide which operator will get evaluated first in a complicated looking compound proposition.

For example: $(p \rightarrow (q \wedge (\neg p)))$

$p = \text{true} \qquad q = \text{false}$

$$p \rightarrow q \wedge \neg p$$

$$T \quad \underbrace{F \wedge F}_{F} \quad = F$$

Operators	Names	Precedence
$\neg$	Negation	1
$\wedge$	Conjunction	2
$\vee$	Disjunction	3
$\rightarrow$	Implication	4
$\leftrightarrow$	Biconditional	5

Problem: Are these system specifications consistent?

"The system is in multiuser state if and only if it is operating normally."

"If the system is operating normally, then the kernel is functioning."

"The kernel is not functioning or the system is in interrupt mode."

"If the system is not in multiuser state, then it is in interrupt mode."

"The system is not in interrupt mode."

Solution: Let p = "The system is in multiuser state."

q = "The system is operating normally."

r = "The kernel is functioning."

s = "The system is in interrupt mode."

1. $p \leftrightarrow q$
2. $q \rightarrow r$
3. $\neg r \vee s$
4. $\neg p \rightarrow s$
5. $\neg s$

Consistent means assigning truth values to propositional variables in such a way that finally we would be able to make all specifications "true".

"The Result of the toss is head if and only if I am saying the truth."

Which of the following options is correct?

(a) The result is head.

(b) The result is tail.

(c) If the person is of Type 2, then the result is tail.

(d) If the person is of Type 1, then the result is tail.

[GATE 2015 (Set - 3): 1 Mark]

Solution: Let p = result of toss is head and q = I am saying the truth

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

CASE 1: Type 1 person

$p \leftrightarrow q = T$

① $p = T \quad q = T$

× ② $p = F \quad q = F$

Result = Head

CASE 2: Type 2 person

$p \leftrightarrow q = F$

① $p = T \quad q = F$

× ② $p = F \quad q = T$

Result = Head

Result of toss is Head

Tautology, Contradiction, Contingency and Satisfiability

Tautology: Compound proposition which is always TRUE

Example:

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction: Compound proposition which is always FALSE.

Example:

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

Contingency: Compound proposition which is sometimes TRUE and sometimes FALSE.

Example:

p	q	p^q
T	T	T
T	F	F
F	T	F
F	F	F

Satisfiability: A compound proposition is satisfiable if there is at least one TRUE result in its truth table.

Unsatisfiability: not even a single TRUE result in its truth table.

Valid: A compound proposition is valid when it is a tautology.

Invalid: A compound proposition is invalid when it is either a contradiction or contingency.

Logical Equivalences

Definition: The compound propositions p and q are said to be logically equivalent if $p \leftrightarrow q$ is a Tautology. Logical Equivalence is denoted by $\equiv$ or $\Leftrightarrow$

Example: $p \wedge T \equiv p$

p	T	P ^ T
T	T	T
F	T	F

Most common and famous Logical Equivalences:

1. Identity Laws: (a) $p \wedge T \equiv p$ (b) $p \vee F \equiv p$
2. Domination Laws: (a) $p \vee T \equiv T$ (b) $p \wedge F \equiv F$
3. Idempotent Laws: (a) $p \vee p \equiv p$ (b) $p \wedge p \equiv p$
4. Double Negation Law: $\neg(\neg p) \equiv p$
5. Commutative Laws: (a) $p \vee q \equiv q \vee p$ (b) $p \wedge q \equiv q \wedge p$
6. Associative Laws: (a) $(p \vee q) \vee r \equiv p \vee (q \vee r)$ (b) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7. Distributive Laws: (a) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ (b) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8. De Morgan's Laws: (a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (b) $\neg(p \vee q) \equiv \neg p \wedge \neg q$
9. Absorption Laws: (a) $p \vee (p \wedge q) \equiv p$ (b) $p \wedge (p \vee q) \equiv p$
10. Negation Laws: (a) $p \vee \neg p \equiv T$ (b) $p \wedge \neg p \equiv F$

Logical Equivalences involving conditional statements:

1. $p \to q \equiv \neg p \vee q$ (Imp)

F T

T T

$T \to F$ $F \vee F = F$

$= F$

2. $p \to q \equiv \neg q \to \neg p$

3. $p \vee q \equiv \neg p \to q$

$\neg(\neg p) \vee q = p \vee q$

4. $p \wedge q \equiv \neg(q \to \neg p)$

$\neg(\neg q \vee \neg p)$

$= q \wedge p$

$= p \wedge q$

5. $\neg(p \to q) \equiv p \wedge \neg q$

6. $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

$(\neg p \vee q) \wedge (\neg p \vee r)$

$= \neg p \vee \underbrace{(q \wedge r)}_{s}$

$= \neg p \vee s = p \to s$

$= p \to (q \wedge r)$

7. $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$

8. $(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$

9. $(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Logical Equivalences involving biconditionals:

1. $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

$$= (p \to q) \wedge (q \to p)$$

$$= (\neg q \to \neg p) \wedge (\neg p \to \neg q)$$

$$= (\underbrace{\neg p}_{s} \to \underbrace{\neg q}_{r}) \wedge (\neg q \to \neg p)$$

$$= (s \to r) \wedge (r \to s) = s \leftrightarrow r$$

$$= \neg p \leftrightarrow \neg q$$

3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

$$= (p \to q) \wedge (q \to p)$$

$$= (\neg p \vee q) \wedge (\neg q \vee p) \qquad (a+b)(c+d) = ac + ad + bc + bd$$

$$= (\neg p \wedge \neg q) \vee \underbrace{(\neg p \wedge p)}_{F} \vee \underbrace{(q \wedge \neg q)}_{F} \vee (q \wedge p)$$

$$= (\neg p \wedge \neg q) \vee F \vee F \vee (q \wedge p) = (\neg p \wedge \neg q) \vee \underbrace{(q \wedge p)}_{p \wedge q}$$

4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

1. $p \to q \equiv \neg p \vee q$

- **Set Theory Translation:**

$$A \subseteq B \equiv A^c \cup B$$

Where:

- $A \subseteq B$: Every element of A is also in B.
- A^c: Complement of A.
- $\cup$: Union of sets.

2. $p \to q \equiv \neg q \to \neg p$

- **Set Theory Translation** (Contrapositive Rule):

$$A \subseteq B \equiv B^c \subseteq A^c$$

Where:

- $B^c \subseteq A^c$: The complement of B is a subset of the complement of A.

4. $p \wedge q \equiv \neg(p \to \neg q)$

- **Set Theory Translation:**

$$A \cap B \equiv (A \subseteq B^c)^c$$

Where:

- $A \cap B$: Intersection of A and B.
- $A \subseteq B^c$: A is a subset of the complement of B.
- $(\cdot)^c$: Complement of the statement.

5. $\neg(p \to q) \equiv p \wedge \neg q$

- **Set Theory Translation:**

$$(A \subseteq B)^c \equiv A \cap B^c$$

Where:

- $A \cap B^c$: Elements in A but not in B.

6. $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$

- **Set Theory Translation:**

$$(A \subseteq B) \cap (A \subseteq C) \equiv A \subseteq (B \cap C)$$

Where:

- $(B \cap C)$: Intersection of B and C.

7. $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$

- **Set Theory Translation:**

$$(A \subseteq C) \cap (B \subseteq C) \equiv (A \cup B) \subseteq C$$

Where:

- $(A \cup B)$: Union of A and B.

8. $(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$

- Set Theory Translation:

$$(A \subseteq B) \cup (A \subseteq C) \equiv A \subseteq (B \cup C)$$

Where:

- $(B \cup C)$: Union of B and C.

TABLE 7 Logical Equivalences Involving Conditional Statements.

$p \to q \equiv \neg p \lor q$
$p \to q \equiv \neg q \to \neg p$
$p \lor q \equiv \neg p \to q$
$p \land q \equiv \neg(p \to \neg q)$
$\neg(p \to q) \equiv p \land \neg q$
$(p \to q) \land (p \to r) \equiv p \to (q \land r)$
$(p \to r) \land (q \to r) \equiv (p \lor q) \to r$
$(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$
$(p \to r) \lor (q \to r) \equiv (p \land q) \to r$

9. $(p \lor q) \to r \equiv (p \to r) \land (q \to r)$

- Set Theory Translation:

$$(A \cup B) \subseteq C \equiv (A \subseteq C) \cap (B \subseteq C)$$

Where:

- $A \cup B$: Union of A and B.
- $\cap$: Intersection of subsets.

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

1. $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$

- **Set Theory Translation:**

$$A = B \equiv (A \subseteq B) \cap (B \subseteq A)$$

Where:

- $A = B$: Equality of sets A and B (i.e., they contain exactly the same elements).
- $A \subseteq B$: A is a subset of B.
- $B \subseteq A$: B is a subset of A.
- $\cap$: Intersection of the subset conditions.

2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

- **Set Theory Translation:**

$$A = B \equiv A^c = B^c$$

Where:

- A^c: Complement of set A.
- B^c: Complement of set B.
- The equivalence indicates that if two sets A and B are equal, their complements are also equal.

3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

- **Set Theory Translation:**

$$A = B \equiv (A \cap B) \cup (A^c \cap B^c)$$

Where:

- $A \cap B$: Intersection of A and B (elements common to both sets).
- $A^c \cap B^c$: Intersection of the complements of A and B (elements not in A and not in B).
- $\cup$: Union of the two conditions.

4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

- **Set Theory Translation:**

$$(A \neq B) \equiv A = B^c$$

Where:

- $A \neq B$: The sets A and B are not equal.
- $A = B^c$: A is equal to the complement of B.

Inference

What is an Inference?

Types of Inferences

(Deductive, Inductive, Abductive)

Examples of Inference

What is an Inference?

An inference is a conclusion reached on the basis of evidence and reasoning.

Evidence+Knowledge=Inference

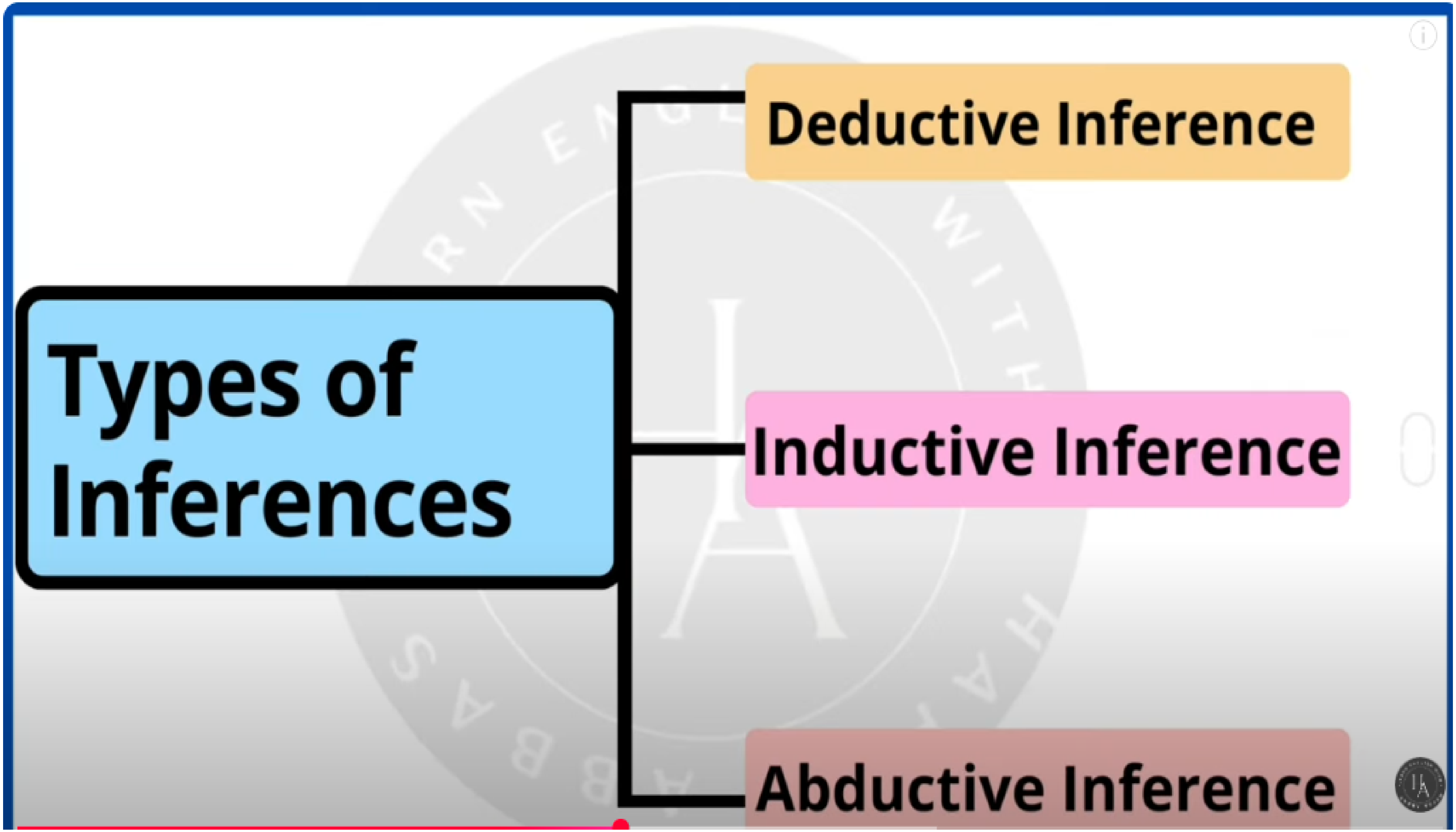
Examples of Inference

Stop

Ready

Go

He doesn't like it


Types of Inferences
Deductive Inference
Inductive Inference
Abductive Inference

DUCTION MEANS GUIDANCE		
DEDUCTIVE	INDUCTIVE	ABDUCTIVE
DE means from	In means towards	Ab Means Away
TOP-Down -logic	Bottom-Up-Logic	
Deductive inferences uses available facts, Information, or Knowledge to deduce a valid Conculsion	Inductive inferences, or inductive Logic, is a type of reasoning that involves drawing a general conculsion from a set of specific observations	Abdductive inferencesis the method of reasoning that involves inferring which of several explanations for particular observed facts is the most completing one.
Example: All dogs have ears, Golden retrievers are dogs, therefore they have ears.	Example: All the swans I have been seen are white.(Premise) Therefore most swans are probably white (Conculsion)	Example: The grass is wet (the observation), therefore, it probably rained last night (The most likely hypothesis)

Rules of Inference in Propositional Logic
(Basic Terminology and Examples)

Premise: is a proposition on the basis of which we would able to draw a conclusion. You can think of premise as an evidence or assumption.

Therefore, initially we assume something is true and on the basis of that assumption, we draw some conclusion.

Conclusion: is a proposition that is reached from the given set of premises. You can think of it as the result of the assumptions that we made in an argument.

if premise then conclusion

Argument: Sequence of statements that ends with a conclusion.

OR

it is a set of one or more premises and a conclusion.

Valid Argument: an argument is said to be valid if and only if it is not possible to make all premises true and a conclusion false.

Example of an argument:

P_1	"If I love cat then I love dog."	$\overset{T}{p} \rightarrow \overset{T}{q}$	T	$((p \rightarrow q) \wedge p) \rightarrow q$
P_2	"I love cat."	p	T	
C	Therefore, "I love dog."	$\therefore q$	T	valid Argument.

"If I love cat then I love dog."	$\overset{F}{p} \rightarrow \overset{T}{q}$	T	Invalid Argument
"I love dog."	q	T	
Therefore, "I love cat."	$\therefore p$	F	

Types of Inference Rules:

1. Modus Ponens: (Law of Detachment)

$$\begin{array}{l} p \to q \\ \underline{p} \\ \therefore q \end{array} \quad \text{OR} \quad [(p \to q) \wedge p] \to q$$

2. Modus Tollens: **(Law of Contrapositive)**

$$\begin{array}{l} p \to q \\ \underline{\sim q} \\ \therefore \sim p \end{array} \quad \text{OR} \quad [(p \to q) \wedge \sim q] \to \sim p$$

3. Hypothetical Syllogism

$$\begin{array}{l} p \to q \\ \underline{q \to r} \\ \therefore p \to r \end{array} \quad \text{OR} \quad [(p \to q) \wedge (q \to r)] \to (p \to r)$$

4. Disjunctive Syllogism

$$\begin{array}{l} p \vee q \\ \underline{\sim p} \\ \therefore q \end{array} \quad \text{OR} \quad [(p \vee q) \wedge \sim p] \to q$$

6. Simplification

$$\frac{p \wedge q}{\therefore p} \quad \text{OR} \quad \frac{p \wedge q}{\therefore q} \quad \text{OR} \quad (p \wedge q) \rightarrow p \quad \text{or} \quad (p \wedge q$$

7. Conjunction

$$\begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array} \quad \text{OR} \quad [(p) \wedge (q)] \rightarrow (p \wedge q)$$

8. Resolution

$$\begin{array}{l} p \vee q \\ \underline{\sim p \vee r} \\ \therefore q \vee r \end{array} \quad \text{OR} \quad [(p \vee q) \wedge (\sim p \vee r)] \rightarrow (q \vee r)$$

Rule of Inference Table

Rule of Inference	Description
Modus Ponens (MP)	If P implies Q, and P is true, then Q is true.
Modus Tollens (MT)	If P implies Q, and Q is false, then P is false.
Hypothetical Syllogism (HS)	If P implies Q and Q implies R, then P implies R.
Disjunctive Syllogism (DS)	If P or Q is true, and P is false, then Q is true.
Addition (Add)	If P is true, then P or Q is true.
Simplification (Simp)	If P and Q are true, then P is true
Conjunction (Conj)	If P and Q are true, then P and Q are true.

Rules of Inference for Propositional Logic

(Definition and Types of Inference Rules)

Rules of Inference: are the templates for constructing valid arguments

deriving conclusions from evidences.

Types of Inference Rules:

1. Modus Ponens:

$$\begin{array}{ll} \overset{T}{p} \rightarrow \overset{T}{q} & T \\ p & T \\ \hline \therefore q \; T & \end{array} \qquad \text{OR} \qquad [(p \rightarrow q) \wedge p] \rightarrow q$$

2. Modus Tollens:

$$\begin{array}{ll} \overset{F}{p} \rightarrow \overset{F}{q} & T \\ \sim q & T \\ \hline \therefore \sim p \; T & \end{array} \qquad \text{OR} \qquad [(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

3. Hypothetical Syllogism

$$\begin{array}{l} p \to q \\ \underline{q \to r} \\ \therefore p \to r \end{array}$$

OR

$$[(p \to q) \wedge (q \to r)] \to (p \to r)$$

4. Disjunctive Syllogism

$$\begin{array}{l} p \vee q \\ \underline{\sim p} \\ \therefore q \end{array}$$

OR

$$[(p \vee q) \wedge \sim p] \to q$$

5. Addition

$$\begin{array}{l} \underline{p} \\ \therefore p \vee q \end{array}$$

OR

$$p \to (p \vee q)$$

6. Simplification

$$\begin{array}{ll} \overset{T}{p} \wedge \overset{T}{q} & T \\ \hline \therefore p & T \end{array} \quad \text{OR} \quad \begin{array}{ll} p \wedge q & \\ \hline \therefore q & T \end{array} \quad \text{OR} \quad (p \wedge q) \to p \quad \text{or} \quad (p \wedge q) \to q$$

7. Conjunction

$$\begin{array}{ll} p & T \\ q & T \\ \hline \therefore \underset{T}{p} \wedge \underset{T}{q} & = T \end{array} \quad \text{OR} \quad [(p) \wedge (q)] \to (p \wedge q)$$

8. Resolution

$$\begin{array}{lll} T/F & \overset{T/F}{p} \vee \overset{T}{q} & T \\ F/T & \sim p \vee \overset{T}{r} & T \\ \hline & \therefore q \vee r & = T \end{array} \quad \text{OR} \quad [(p \vee q) \wedge (\sim p \vee r)] \to (q \vee r)$$

1. Modus Ponens: (Law of Detachment)

$$\begin{array}{l} p \to q \\ \underline{p} \\ \therefore q \end{array} \qquad \text{OR} \qquad [(p \to q) \wedge p] \to q$$

Example 1:

Premise 1: If it rains, the ground will be wet. (P → Q)

Premise 2: It is raining. (P)

Conclusion: The ground will be wet. (Q)

Example 2:

Premise 1: If a person studies hard, they will pass the exam. (P → Q)

Premise 2: This person studied hard. (P)

Conclusion: They will pass the exam. (Q)

Example 3:

Premise 1: If you eat too much sugar, you will get a cavity. (P → Q)

Premise 2: You ate too much sugar. (P)

Conclusion: You will get a cavity. (Q)

Example 4:

Premise 1: If a number is divisible by 4, it is also divisible by 2. (P → Q)

Premise 2: 16 is divisible by 4. (P)

Conclusion: 16 is divisible by 2. (Q)

2. Modus Tollens: (Law of Contrapositive)

$$\begin{array}{l} p \rightarrow q \\ \sim q \\ \hline \therefore \sim p \end{array} \qquad \text{OR} \qquad [(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$$

Example 1:

Premise 1: If it rains, the ground will be wet. (P → Q)

Premise 2: The ground is not wet. (¬Q)

Conclusion: It is not raining. (¬P)

Example 2:

Premise 1: If a person studies hard, they will pass the exam. (P → Q)

Premise 2: This person did not pass the exam. (¬Q)

Conclusion: They did not study hard. (¬P)

Example 3:

Premise 1: If a car has enough fuel, it will start. (P → Q)

Premise 2: The car did not start. (¬Q)

Conclusion: The car did not have enough fuel. (¬P)

Example 4:

Premise 1: If a plant gets sunlight, it will grow. (P → Q)

Premise 2: The plant did not grow. (¬Q)

Conclusion: The plant did not get sunlight. (¬P)

3. Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

OR

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

Example 1:

Premise 1: If it rains, the ground will be wet. $(P \rightarrow Q)$

Premise 2: If the ground is wet, the grass will grow. $(Q \rightarrow R)$

Conclusion: If it rains, the grass will grow. $(P \rightarrow R)$

Example 2:

Premise 1: If a person studies hard, they will pass the exam. $(P \rightarrow Q)$

Premise 2: If they pass the exam, they will get a good job. $(Q \rightarrow R)$

Conclusion: If a person studies hard, they will get a good job. $(P \rightarrow R)$

Example 3:

Premise 1: If a car has fuel, it will start. $(P \rightarrow Q)$

Premise 2: If a car starts, it will run. $(Q \rightarrow R)$

Conclusion: If a car has fuel, it will run. $(P \rightarrow R)$

Example 4:

Premise 1: If someone exercises, they will stay healthy. $(P \rightarrow Q)$

Premise 2: If they stay healthy, they will live longer. $(Q \rightarrow R)$

Conclusion: If someone exercises, they will live longer. $(P \rightarrow R)$

4. Disjunctive Syllogism

$$\begin{array}{l} p \vee q \\ \underline{\sim p} \\ \therefore q \end{array} \quad \text{OR} \quad [(p \vee q) \wedge \sim p] \rightarrow q$$

Example 1:

Premise 1: Either the light is on, or the room is dark. (P ∨ Q)

Premise 2: The light is not on. (¬P)

Conclusion: The room is dark. (Q)

Example 2:

Premise 1: Either I will go for a walk, or I will stay home. (P ∨ Q)

Premise 2: I will not go for a walk. (¬P)

Conclusion: I will stay home. (Q)

Example 3:

Premise 1: Either the food is spoiled, or it is safe to eat. (P ∨ Q)

Premise 2: The food is not spoiled. (¬P)

Conclusion: It is safe to eat. (Q)

Example 4:

Premise 1: Either John will pass the exam, or Mary will tutor him. (P ∨ Q)

Premise 2: John will not pass the exam. (¬P)

Conclusion: Mary will tutor him. (Q)

6. Simplification $\dfrac{p \wedge q}{\therefore p}$ OR $\dfrac{p \wedge q}{\therefore q}$ OR $(p \wedge q) \to p$ or $(p \wedge q$

Example 1:
Premise: It is raining, and the ground is wet. (P ∧ Q)
Conclusion 1: It is raining. (P)
Conclusion 2: The ground is wet. (Q)

Example 2:
Premise: John is studying, and Mary is cooking. (P ∧ Q)
Conclusion 1: John is studying. (P)
Conclusion 2: Mary is cooking. (Q)

Example 3:
Premise: The car has fuel, and the engine is running. (P ∧ Q)
Conclusion 1: The car has fuel. (P)
Conclusion 2: The engine is running. (Q)

Example 4:
Premise: The package arrived, and the recipient signed for it. (P ∧ Q)
Conclusion 1: The package arrived. (P)
Conclusion 2: The recipient signed for it. (Q)

7. Conjunction

$$\begin{array}{l} p \\ \underline{q} \\ \therefore p \wedge q \end{array}$$

OR

$$[(p) \wedge (q)] \to (p \wedge q)$$

Example 1:

Premise 1: It is sunny. (P)

Premise 2: It is warm. (Q)

Conclusion: It is sunny and warm. (P ∧ Q)

Example 2:

Premise 1: The car has fuel. (P)

Premise 2: The car engine is running. (Q)

Conclusion: The car has fuel, and the car engine is running. (P ∧ Q)

Example 3:

Premise 1: John is studying. (P)

Premise 2: Mary is cooking. (Q)

Conclusion: John is studying, and Mary is cooking. (P ∧ Q)

Example 4:

Premise 1: The package arrived. (P)

Premise 2: The recipient signed for it. (Q)

Conclusion: The package arrived, and the recipient signed for it. (P ∧ Q)

8. Resolution

$$\begin{array}{l} p \lor q \\ \underline{\sim p \lor r} \\ \therefore q \lor r \end{array} \quad \text{OR} \quad [(p \lor q) \land (\sim p \lor r)] \rightarrow (q \lor r)$$

Example 1:

Premise 1: The light is on, or the room is dark. (P ∨ Q)

Premise 2: The light is not on, or the window is open. (¬P ∨ R)

Conclusion: The room is dark, or the window is open. (Q ∨ R)

Example 2:

Premise 1: John studies hard, or he fails the exam. (P ∨ Q)

Premise 2: John does not study hard, or he gets a good grade. (¬P ∨ R)

Conclusion: He fails the exam, or he gets a good grade. (Q ∨ R)

Example 3:

Premise 1: The package arrived, or it got lost. (P ∨ Q)

Premise 2: The package did not arrive, or it was delayed. (¬P ∨ R)

Conclusion: It got lost, or it was delayed. (Q ∨ R)

Example 4:

Premise 1: The sky is clear, or it will rain. (P ∨ Q)

Premise 2: The sky is not clear, or it will be windy. (¬P ∨ R)

Conclusion: It will rain, or it will be windy. (Q ∨ R)

1. Which of these sentences are propositions? What are the truth values of those that are propositions?

a) Boston is the capital of Massachusetts.

b) Miami is the capital of Florida.

c) $2 + 3 = 5$.

d) $5 + 7 = 10$.

e) $x + 2 = 11$.

f) Answer this question.

SOLUTIONS

1. Propositions must have clearly defined truth values, so a proposition must be a declarative sentence with no free variables.

a) This is a true proposition.

b) This is a false proposition (Tallahassee is the capital).

c) This is a true proposition.

d) This is a false proposition.

e) This is not a proposition (it contains a variable; the truth value depends on the value assigned to x).

f) This is not a proposition, since it does not assert anything.

3. What is the negation of each of these propositions?

a) Mei has an MP3 player.

b) There is no pollution in New Jersey.

c) $2 + 1 = 3$.

d) The summer in Maine is hot and sunny.

SOLUTION

3. a) Mei does not have an MP3 player.

b) There is pollution in New Jersey.

c) $2 + 1 \neq 3$

d) It is not the case that the summer in Maine is hot and sunny. In other words, the summer in Maine is not hot and sunny, which means that it is not hot *or* it is not sunny. It is not correct to negate this by saying "The summer in Maine is not hot and not sunny." [For this part (and in a similar vein for part (**b**)) we need to assume that there are well-defined notions of hot and sunny; otherwise this would not be a proposition because of not having a definite truth value.]

9. Let p and q be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.

a) $\neg q$ **b)** $p \wedge q$ **c)** $\neg p \vee q$
d) $p \to \neg q$ **e)** $\neg q \to p$ **f)** $\neg p \to \neg q$
g) $p \leftrightarrow \neg q$ **h)** $\neg p \wedge (p \vee \neg q)$

9. Let p and q be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.

a) $\neg q$ **b)** $p \wedge q$ **c)** $\neg p \vee q$
d) $p \to \neg q$ **e)** $\neg q \to p$ **f)** $\neg p \to \neg q$
g) $p \leftrightarrow \neg q$ **h)** $\neg p \wedge (p \vee \neg q)$

9. This is pretty straightforward, using the normal words for the logical operators.

a) Sharks have not been spotted near the shore.

b) Swimming at the New Jersey shore is allowed, and sharks have been spotted near the shore.

c) Swimming at the New Jersey shore is not allowed, or sharks have been spotted near the shore.

d) If swimming at the New Jersey shore is allowed, then sharks have not been spotted near the shore.

e) If sharks have not been spotted near the shore, then swimming at the New Jersey shore is allowed.

f) If swimming at the New Jersey shore is not allowed, then sharks have not been spotted near the shore.

g) Swimming at the New Jersey shore is allowed if and only if sharks have not been spotted near the shore.

h) Swimming at the New Jersey shore is not allowed, and either swimming at the New Jersey shore is allowed or sharks have not been spotted near the shore. Note that we were able to incorporate the parentheses by using the word "either" in the second half of the sentence.

11. Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

a) It is below freezing and snowing.

b) It is below freezing but not snowing.

c) It is not below freezing and it is not snowing.

d) It is either snowing or below freezing (or both).

e) If it is below freezing, it is also snowing.

f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.

g) That it is below freezing is necessary and sufficient for it to be snowing.

11. Let p and q be the propositions

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

a) It is below freezing and snowing.
b) It is below freezing but not snowing.
c) It is not below freezing and it is not snowing.
d) It is either snowing or below freezing (or both).
e) If it is below freezing, it is also snowing.
f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
g) That it is below freezing is necessary and sufficient for it to be snowing.

11. **a)** Here we have the conjunction $p \wedge q$.

b) Here we have a conjunction of p with the negation of q, namely $p \wedge \neg q$. Note that "but" logically means the same thing as "and."

c) Again this is a conjunction: $\neg p \wedge \neg q$.

d) Here we have a disjunction, $p \vee q$. Note that $\vee$ is the inclusive *or*, so the "(or both)" part of the English sentence is automatically included.

e) This sentence is a conditional statement, $p \rightarrow q$.

f) This is a conjunction of propositions, both of which are compound: $(p \vee q) \wedge (p \rightarrow \neg q)$.

g) This is the biconditional $p \leftrightarrow q$.

13. Let p and q be the propositions

p : You drive over 65 miles per hour.

q : You get a speeding ticket.

Write these propositions using p and q and logical connectives (including negations).

a) You do not drive over 65 miles per hour.

b) You drive over 65 miles per hour, but you do not get a speeding ticket.

c) You will get a speeding ticket if you drive over 65 miles per hour.

d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.

e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.

f) You get a speeding ticket, but you do not drive over 65 miles per hour.

g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

13. Let p and q be the propositions

p : You drive over 65 miles per hour.

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a) You do not drive over 65 miles per hour.

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e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.

f) You get a speeding ticket, but you do not drive over 65 miles per hour.

g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

13. **a)** This is just the negation of p, so we write $\neg p$.

b) This is a conjunction ("but" means "and"): $p \wedge \neg q$.

c) The position of the word "if" tells us which is the antecedent and which is the consequence: $p \rightarrow q$.

d) $\neg p \rightarrow \neg q$

e) The sufficient condition is the antecedent: $p \rightarrow q$.

f) $q \wedge \neg p$

g) "Whenever" means "if": $q \rightarrow p$.

19. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a) Coffee or tea comes with dinner.

b) A password must have at least three digits or be at least eight characters long.

c) The prerequisite for the course is a course in number theory or a course in cryptography.

d) You can pay using U.S. dollars or euros.

19. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

a) Coffee or tea comes with dinner.

b) A password must have at least three digits or be at least eight characters long.

c) The prerequisite for the course is a course in number theory or a course in cryptography.

d) You can pay using U.S. dollars or euros.

19. **a)** Presumably the diner gets to choose only one of these beverages, so this is an exclusive *or*.

b) This is probably meant to be inclusive, so that long passwords with many digits are acceptable.

c) This is surely meant to be inclusive. If a student has had both of the prerequisites, so much the better.

d) At first glance one might argue that no one would pay with both currencies simultaneously, so it would seem reasonable to call this an exclusive *or*. There certainly could be cases, however, in which the patron would pay a portion of the bill in dollars and the remainder in euros. Therefore, an inclusive *or* seems better.

25. Write each of these propositions in the form "p if and only if q" in English.

a) If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.

b) For you to win the contest it is necessary and sufficient that you have the only winning ticket.

c) You get promoted only if you have connections, and you have connections only if you get promoted.

d) If you watch television your mind will decay, and conversely.

e) The trains run late on exactly those days when I take it.

25. Write each of these propositions in the form "p if and only if q" in English.

a) If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.

b) For you to win the contest it is necessary and sufficient that you have the only winning ticket.

c) You get promoted only if you have connections, and you have connections only if you get promoted.

d) If you watch television your mind will decay, and conversely.

e) The trains run late on exactly those days when I take it.

25. In each case there will be two statements. It is being asserted that the first one holds true if and only if the second one does. The order doesn't matter, but often one order is more colloquial English.

a) You buy an ice cream cone if and only if it is hot outside.

b) You win the contest if and only if you hold the only winning ticket.

c) You get promoted if and only if you have connections.

d) Your mind will decay if and only if you watch television.

e) The train runs late if and only if it is a day I take the train.

Exercises

1. Which of these sentences are propositions? What are the truth values of those that are propositions?

a) Boston is the capital of Massachusetts.

b) Miami is the capital of Florida.

c) $2 + 3 = 5$.

d) $5 + 7 = 10$.

e) $x + 2 = 11$.

f) Answer this question.

2. Which of these are propositions? What are the truth values of those that are propositions?

a) Do not pass go.

b) What time is it?

c) There are no black flies in Maine.

d) $4 + x = 5$.

e) The moon is made of green cheese.

f) $2^n \geq 100$.

3. What is the negation of each of these propositions?

a) Mei has an MP3 player.

b) There is no pollution in New Jersey.

c) $2 + 1 = 3$.

d) The summer in Maine is hot and sunny.

4. What is the negation of each of these propositions?

a) Jennifer and Teja are friends.

b) There are 13 items in a baker's dozen.

c) Abby sent more than 100 text messages every day.

d) 121 is a perfect square.

5. What is the negation of each of these propositions?

a) Steve has more than 100 GB free disk space on his laptop.
b) Zach blocks e-mails and texts from Jennifer.
c) $7 \cdot 11 \cdot 13 = 999$.
d) Diane rode her bicycle 100 miles on Sunday.

6. Suppose that Smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions.

a) Smartphone B has the most RAM of these three smartphones.
b) Smartphone C has more ROM or a higher resolution camera than Smartphone B.
c) Smartphone B has more RAM, more ROM, and a higher resolution camera than Smartphone A.
d) If Smartphone B has more RAM and more ROM than Smartphone C, then it also has a higher resolution camera.
e) Smartphone A has more RAM than Smartphone B if and only if Smartphone B has more RAM than Smartphone A.

7. Suppose that during the most recent fiscal year, the annual revenue of Acme Computer was 138 billion dollars and its net profit was 8 billion dollars, the annual revenue of Nadir Software was 87 billion dollars and its net profit was 5 billion dollars, and the annual revenue of Quixote Media was 111 billion dollars and its net profit was 13 billion dollars. Determine the truth value of each of these propositions for the most recent fiscal year.

a) Quixote Media had the largest annual revenue.
b) Nadir Software had the lowest net profit and Acme Computer had the largest annual revenue.
c) Acme Computer had the largest net profit or Quixote Media had the largest net profit.
d) If Quixote Media had the smallest net profit, then Acme Computer had the largest annual revenue.
e) Nadir Software had the smallest net profit if and only if Acme Computer had the largest annual revenue.

8. Let p and q be the propositions

p : I bought a lottery ticket this week.
q : I won the million dollar jackpot.

Express each of these propositions as an English sentence.

a) $\neg p$ **b)** $p \vee q$ **c)** $p \rightarrow q$
d) $p \wedge q$ **e)** $p \leftrightarrow q$ **f)** $\neg p \rightarrow \neg q$
g) $\neg p \wedge \neg q$ **h)** $\neg p \vee (p \wedge q)$

9. Let p and q be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore," respectively. Express each of these compound propositions as an English sentence.

a) $\neg q$ **b)** $p \wedge q$ **c)** $\neg p \vee q$
d) $p \rightarrow \neg q$ **e)** $\neg q \rightarrow p$ **f)** $\neg p \rightarrow \neg q$
g) $p \leftrightarrow \neg q$ **h)** $\neg p \wedge (p \vee \neg q)$

10. Let p and q be the propositions "The election is decided" and "The votes have been counted," respectively. Express each of these compound propositions as an English sentence.

a) $\neg p$ **b)** $p \vee q$
c) $\neg p \wedge q$ **d)** $q \rightarrow p$
e) $\neg q \rightarrow \neg p$ **f)** $\neg p \rightarrow \neg q$
g) $p \leftrightarrow q$ **h)** $\neg q \vee (\neg p \wedge q)$

11. Let p and q be the propositions

p : It is below freezing.
q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

a) It is below freezing and snowing.
b) It is below freezing but not snowing.
c) It is not below freezing and it is not snowing.
d) It is either snowing or below freezing (or both).
e) If it is below freezing, it is also snowing.
f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
g) That it is below freezing is necessary and sufficient for it to be snowing.

12. Let p, q, and r be the propositions

p : You have the flu.
q : You miss the final examination.
r : You pass the course.

Express each of these propositions as an English sentence.

a) $p \rightarrow q$ **b)** $\neg q \leftrightarrow r$
c) $q \rightarrow \neg r$ **d)** $p \vee q \vee r$
e) $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$
f) $(p \wedge q) \vee (\neg q \wedge r)$

13. Let p and q be the propositions

p : You drive over 65 miles per hour.
q : You get a speeding ticket.

Write these propositions using p and q and logical connectives (including negations).

a) You do not drive over 65 miles per hour.
b) You drive over 65 miles per hour, but you do not get a speeding ticket.
c) You will get a speeding ticket if you drive over 65 miles per hour.
d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.
e) Driving over 65 miles per hour is sufficient for getting a speeding ticket.
f) You get a speeding ticket, but you do not drive over 65 miles per hour.
g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.

14. Let p, q, and r be the propositions

p : You get an A on the final exam.
q : You do every exercise in this book.
r : You get an A in this class.

Write these propositions using p, q, and r and logical connectives (including negations).

a) You get an A in this class, but you do not do every exercise in this book.
b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
c) To get an A in this class, it is necessary for you to get an A on the final.
d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

15. Let p, q, and r be the propositions

p : Grizzly bears have been seen in the area.
q : Hiking is safe on the trail.
r : Berries are ripe along the trail.

Write these propositions using p, q, and r and logical connectives (including negations).

a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.
b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail.
c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.
d) It is not safe to hike on the trail, but grizzly bears have not been seen in the area and the berries along the trail are ripe.
e) For hiking on the trail to be safe, it is necessary but not sufficient that berries not be ripe along the trail and for grizzly bears not to have been seen in the area.
f) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

16. Determine whether these biconditionals are true or false.

a) $2+2=4$ if and only if $1+1=2$.
b) $1+1=2$ if and only if $2+3=4$.
c) $1+1=3$ if and only if monkeys can fly.
d) $0>1$ if and only if $2>1$.

17. Determine whether each of these conditional statements is true or false.

a) If $1+1=2$, then $2+2=5$.
b) If $1+1=3$, then $2+2=4$.
c) If $1+1=3$, then $2+2=5$.
d) If monkeys can fly, then $1+1=3$.

18. Determine whether each of these conditional statements is true or false.

a) If $1+1=3$, then unicorns exist.
b) If $1+1=3$, then dogs can fly.
c) If $1+1=2$, then dogs can fly.
d) If $2+2=4$, then $1+2=3$.

19. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

- **a)** Coffee or tea comes with dinner.
- **b)** A password must have at least three digits or be at least eight characters long.
- **c)** The prerequisite for the course is a course in number theory or a course in cryptography.
- **d)** You can pay using U.S. dollars or euros.

20. For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.

- **a)** Experience with C++ or Java is required.
- **b)** Lunch includes soup or salad.
- **c)** To enter the country you need a passport or a voter registration card.
- **d)** Publish or perish.

21. For each of these sentences, state what the sentence means if the logical connective or is an inclusive or (that is, a disjunction) versus an exclusive or. Which of these meanings of or do you think is intended?

- **a)** To take discrete mathematics, you must have taken calculus or a course in computer science.
- **b)** When you buy a new car from Acme Motor Company, you get \$2000 back in cash or a 2% car loan.
- **c)** Dinner for two includes two items from column A or three items from column B.
- **d)** School is closed if more than 2 feet of snow falls or if the wind chill is below -100.

22. Write each of these statements in the form "if p, then q" in English. [*Hint:* Refer to the list of common ways to express conditional statements provided in this section.]

- **a)** It is necessary to wash the boss's car to get promoted.
- **b)** Winds from the south imply a spring thaw.
- **c)** A sufficient condition for the warranty to be good is that you bought the computer less than a year ago.
- **d)** Willy gets caught whenever he cheats.
- **e)** You can access the website only if you pay a subscription fee.
- **f)** Getting elected follows from knowing the right people.
- **g)** Carol gets seasick whenever she is on a boat.

23. Write each of these statements in the form "if p, then q" in English. [*Hint:* Refer to the list of common ways to express conditional statements.]

- **a)** It snows whenever the wind blows from the northeast.
- **b)** The apple trees will bloom if it stays warm for a week.
- **c)** That the Pistons win the championship implies that they beat the Lakers.
- **d)** It is necessary to walk 8 miles to get to the top of Long's Peak.
- **e)** To get tenure as a professor, it is sufficient to be world-famous.
- **f)** If you drive more than 400 miles, you will need to buy gasoline.
- **g)** Your guarantee is good only if you bought your CD player less than 90 days ago.
- **h)** Jan will go swimming unless the water is too cold.

24. Write each of these statements in the form "if p, then q" in English. [*Hint:* Refer to the list of common ways to express conditional statements provided in this section.]

- **a)** I will remember to send you the address only if you send me an e-mail message.
- **b)** To be a citizen of this country, it is sufficient that you were born in the United States.
- **c)** If you keep your textbook, it will be a useful reference in your future courses.
- **d)** The Red Wings will win the Stanley Cup if their goalie plays well.
- **e)** That you get the job implies that you had the best credentials.
- **f)** The beach erodes whenever there is a storm.
- **g)** It is necessary to have a valid password to log on to the server.
- **h)** You will reach the summit unless you begin your climb too late.

25. Write each of these propositions in the form "p if and only if q" in English.

- **a)** If it is hot outside you buy an ice cream cone, and if you buy an ice cream cone it is hot outside.
- **b)** For you to win the contest it is necessary and sufficient that you have the only winning ticket.
- **c)** You get promoted only if you have connections, and you have connections only if you get promoted.
- **d)** If you watch television your mind will decay, and conversely.
- **e)** The trains run late on exactly those days when I take

26. Write each of these propositions in the form "p if and only if q" in English.

- **a)** For you to get an A in this course, it is necessary and sufficient that you learn how to solve discrete mathematics problems.
- **b)** If you read the newspaper every day, you will be informed, and conversely.
- **c)** It rains if it is a weekend day, and it is a weekend day if it rains.
- **d)** You can see the wizard only if the wizard is not in, and the wizard is not in only if you can see him.

27. State the converse, contrapositive, and inverse of each of these conditional statements.

- **a)** If it snows today, I will ski tomorrow.
- **b)** I come to class whenever there is going to be a quiz.
- **c)** A positive integer is a prime only if it has no divisors other than 1 and itself.

28. State the converse, contrapositive, and inverse of each of these conditional statements.

- **a)** If it snows tonight, then I will stay at home.
- **b)** I go to the beach whenever it is a sunny summer day.
- **c)** When I stay up late, it is necessary that I sleep until noon.

How many rows appear in a truth table for each of these compound propositions?

- **a)** $(q \to \neg p) \vee (\neg p \to \neg q)$
- **b)** $(p \vee \neg t) \wedge (p \vee \neg s)$
- **c)** $(p \to r) \vee (\neg s \to \neg t) \vee (\neg u \to v)$
- **d)** $(p \wedge r \wedge s) \vee (q \wedge t) \vee (r \wedge \neg t)$

Construct a truth table for each of these compound propositions.

- **a)** $p \wedge \neg p$
- **b)** $p \vee \neg p$
- **c)** $(p \vee \neg q) \to q$
- **d)** $(p \vee q) \to (p \wedge q)$
- **e)** $(p \to q) \leftrightarrow (\neg q \to \neg p)$
- **f)** $(p \to q) \to (q \to p)$

Construct a truth table for each of these compound propositions.

- **a)** $p \to \neg p$
- **b)** $p \leftrightarrow \neg p$
- **c)** $p \oplus (p \vee q)$
- **d)** $(p \wedge q) \to (p \vee q)$
- **e)** $(q \to \neg p) \leftrightarrow (p \leftrightarrow q)$
- **f)** $(p \leftrightarrow q) \oplus (p \leftrightarrow \neg q)$

Section 1.1

1. a) Yes, T b) Yes, F c) Yes, T d) Yes, F e) No f) No 3. a) Mei does not have an MP3 player. b) There is pollution in New Jersey. c) $2 + 1 \neq 3$. d) The summer in Maine is not hot or it is not sunny. 5. a) Steve does not have more than 100 GB free disk space on his laptop b) Zach does not block e-mails from Jennifer, or he does not block texts from Jennifer c) $7 \cdot 11 \cdot 13 \neq 999$ d) Diane did not ride her bike 100 miles on Sunday 7. a) F b) T c) T d) T e) T 9. a) Sharks have not been spotted near the shore. b) Swimming at the New Jersey shore is allowed, and sharks have been spotted near the shore. c) Swimming at the New Jersey shore is not allowed, or sharks have been spotted near the shore. d) If swimming at the New Jersey shore is allowed, then sharks have not been spotted near the shore. e) If sharks have not been spotted near the shore, then swimming at the New Jersey shore is allowed. f) If swimming at the New Jersey shore is not allowed, then sharks have not been spotted near the shore. g) Swimming at the New Jersey shore is allowed if and only if sharks have not been spotted near the shore. h) Swimming at the New Jersey shore is not allowed, and either swimming at the New Jersey shore is allowed or sharks have not been spotted near the shore. (Note that we were able to incorporate the parentheses by using the word "either" in the second half of the sentence.) 11. a) $p \wedge q$ b) $p \wedge \neg q$ c) $\neg p \wedge \neg q$ d) $p \vee q$ e) $p \rightarrow q$ f) $(p \vee q) \wedge (p \rightarrow \neg q)$ g) $q \leftrightarrow p$ 13. a) $\neg p$ b) $p \wedge \neg q$ c) $p \rightarrow q$ d) $\neg p \rightarrow \neg q$ e) $p \rightarrow q$ f) $q \wedge \neg p$ g) $q \rightarrow p$ 15. a) $r \wedge \neg p$ b) $\neg p \wedge q \wedge r$ c) $r \rightarrow (q \leftrightarrow \neg p)$ d) $\neg q \wedge \neg p \wedge r$ e) $(q \rightarrow (\neg r \wedge \neg p)) \wedge \neg((\neg r \wedge \neg p) \rightarrow q)$ f) $(p \wedge r) \rightarrow \neg q$ 17. a) False b) True c) True d) True 19. a) Exclusive or: You get only one beverage. b) Inclusive or: Long passwords can have any combination of symbols. c) Inclusive or: A student with both courses is even more qualified. d) Either interpretation possible; a traveler might wish to pay with a mixture of the two currencies, or the store may not allow that. 21. a) Inclusive or: It is allowable to take discrete mathematics if you have had calculus or computer science, or both. Exclusive or: It is allowable to take discrete mathematics if you have had calculus or computer science, but not if you have had both. Most likely the inclusive or is intended. b) Inclusive or: You can take the rebate, or you can get a low-interest loan, or you can get both the rebate and a low-interest loan. Exclusive or: You can take the rebate, or you can get a low-interest loan, but you cannot get both the rebate and a low-interest loan. Most likely the exclusive or is intended. c) Inclusive or: You can order two items from column A and none from column B, or three items from column B and none from column A, or five items including two from column A and three from column B. Exclusive or: You can B, but not both. Almost certainly the exclusive or is intended. d) Inclusive or: More than 2 feet of snow or windchill below -100, or both, will close school. Exclusive or: More than 2 feet of snow or windchill below -100, but not both, will close school. Certainly the inclusive or is intended. 23. a) If the wind blows from the northeast, then it snows. b) If it stays warm for a week, then the apple trees will bloom. c) If the Pistons win the championship, then they beat the Lakers. d) If you get to the top of Long's Peak, then you must have walked 8 miles. e) If you are world-famous, then you will get tenure as a professor. f) If you drive more than 400 miles, then you will need to buy gasoline. g) If your guarantee is good, then you must have bought your CD player less than 90 days ago. h) If the water is not too cold, then Jan will go swimming. 25. a) You buy an ice cream cone if and only if it is hot outside. b) You win the contest if and only if you hold the only winning ticket. c) You get promoted if and only if you have connections. d) Your mind will decay if and only if you watch television. e) The train runs late if and only if it is a day I take the train. 27. a) Converse: "I will ski tomorrow only if it snows today." Contrapositive: "If I do not ski tomorrow, then it will not have snowed today." Inverse: "If it does not snow today, then I will not ski tomorrow." b) Converse: "If I come to class, then there will be a quiz." Contrapositive: "If I do not come to class, then there will not be a quiz." Inverse: "If there is not going to be a quiz, then I don't come to class." c) Converse: "A positive integer is a prime if it has no divisors other than 1 and itself." Contrapositive: "If a positive integer has a divisor other than 1 and itself, then it is not prime." Inverse: "If a positive integer is not prime, then it has a divisor other than 1 and itself." 29. a) 2 b) 16 c) 64 d) 16

31. a)

p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

b)

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

c)

p	q	$\neg q$	$p \vee \neg q$	$(p \vee \neg q) \rightarrow q$
T	T	F	T	T
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

d)

p	q	$p \vee q$	$p \wedge q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T